

Method of Differences

Key Points

Consider a sequence $(a_n)_{n=0}^{\infty}$, such that every term can be written in the form

$$a_k = f(k) - f(k+1),$$

Then

$$\sum_{i=0}^N a_i = f(0) - f(N+1)$$

However it may be that the difference is compared with a term further into the sequence so in practice it is best to write out terms to spot the pattern

Basic Skills

Q1. Let $a_n = \frac{1}{n+1} - \frac{1}{n+2}$. Calculate

a) a_0 b) a_1 c) a_2 d) a_3 e) $a_0 + a_1$

f) $a_1 + a_2$ g) $a_2 + a_3$ h) $a_0 + a_1 + a_2$

i) $a_0 + a_1 + a_2 + a_3$

j) What do you think $\sum_{n=0}^{\infty} a_n$ will be

Q2. Let $f(x) = x^2 - 2x + 3$. Calculate $a_n = f(n+1) - f(n)$

Q3. Express $\frac{5x+11}{(x+1)(x+3)}$ as partial fractions

Q4. Express $\frac{3x+5}{(x-3)(2x+1)}$ as partial fractions

Competency Skills

Q5. We want to calculate $\sum_{i=0}^N \frac{1}{(i+1)(i+2)}$ using a method of differences

a) Express $\frac{1}{(i+1)(i+2)}$ in partial fractions

b) Hence use the method of differences to calculate $\sum_{i=0}^N \frac{1}{(i+1)(i+2)}$

Q6. Calculate $\sum_{i=0}^n \frac{2}{(2i+1)(2i+3)}$ using a method of differences

Q7. Calculate $\sum_{r=1}^n \ln\left(\frac{r+1}{r}\right)$ using a method of differences

Q8. Calculate $\sum_{r=1}^n \frac{1}{r(r+1)(r+2)}$ using a method of differences

Advanced Skills

Q9. Show that if $a_i = f(i) - f(i+1)$ then $\sum_{i=0}^N a_i = f(0) - f(N+1)$

Q10. Calculate $\sum_{r=0}^N \frac{r}{(r+1)(r+2)(r+3)}$ using a method of differences.

Q11. a) Can you use the standard method of differences to work out $\sum_{r=5}^n \frac{1}{(r-2)(r+1)}$

b) Can you adjust the method of differences to help work out the sum in part a)

Q12. a) Calculate $\sum_{i=0}^n \frac{2}{4i^2+16i+15}$ using a method of differences.

b) Hence calculate $\sum_{i=0}^{\infty} \frac{2}{4i^2+16i+15}$

Extension

Q13. Find $\sum_{i=0}^{\infty} \frac{8}{8i^3+36i^2+46i+15}$ using a method of differences

Hint: You need to factorise the cubic

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